# Chapter 10
# Pomegranate (*Punica granatum*)—Morphology, Taxonomy, Composition and Health Benefits

**Abstract** Pomegranate, often referred to as a fruit of heaven, finds mention in various holy manuscripts. Globally, India is the largest producer of pomegranates. Pomegranate is a nutrient dense fruit; however, the presence of hard rind limits its consumption. It is a good source of carbohydrates, fibre, proteins, vitamin E, vitamin K, selenium and magnesium. The bioactivities compounds of fruit (including tannins, flavonoids, alkaloids, organic acids, triterpenes and steroids, etc. are responsible for pharmaceutical, pharmacological and medicinal benefits of pomegranate.



| Family | *Punicaceae* |
|---|---|
| Genus: | *Olea* |
| Species: | *Punica granatum* |
| Local Name: | Anaar, Deain |

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## 1 Origin and Habitat

Pomegranate, often referred to as "Fruit of the heaven" is native to Iran and north western Himalayas and has been grown throughout the Mediterranean Asia, Africa and Europe from centuries. Pomegranate finds mention in various religious manuscripts. Pomegranate tree is well suited for growth in temperate as well as sub-tropical regions and requires cool winter and hot summer. The plant is drought tolerant and prefers semi-arid climate for its successful cultivation. A persistent exposure to sub-freezing temperature is unsuitable for propagation of pomegranate tree. Moreover, the temperature of 12 °C or above is favourable for its fruitful thriving. Pomegranate blossoms well at an altitude of 500–1850 m above sea level and requires a continuous supply of 5.3 mm of water per day. Pomegranate prefers deep, loamy soil with 5.5–7.2 pH and salt concentration of 2000–25000 ppm. However, trees can tolerate sandy soils with moderate pH (Aulakh & Sur, 1999).

## 2 Morphological Characteristics

Pomegranate tree is a member of family "Punicaceae" which comprises only one genus "Punica" and two species. The plant is a deciduous, evergreen attractive shrub 6–10 m in height and long lived. The tree bears thorny multiple branches. The leaves of the plant are evergreen, 1–10 cm long and short stemmed are placed as a cluster of 5–6 on branches. The flowers, 3 cm wide, are characterized by red colour, pointed sepals and numerous stamens. Two to seven flowers are present at the periphery of each branch. The hexagonal-shaped fruit is 6–12 cm wide and weighs around 200 g. The fruit is crowned by thick tubular calyx. The rind of the fruit is red, hard and leathery. The edible fruit, sarcotesta, is a berry which comprises of 600 arils (a casing for seeds) under a thick skin. The ripe fruit attains red colour. The arils are placed in several bunches separated by membranous walls and white spongy tissue (Ataseven Isık & Celik, 2006).

## 3 Production, Taxonomy and Consumption

India with 2.7 million metric tonnes of pomegranate from an area of 22 thousand hectares is the largest producer of pomegranate (Kushwaha et al., 2013) while China and Turkey with 0.3 million metric tonnes and 0.22 million metric tonnes, respectively stands at second and third position in worldwide pomegranate production. Over the years, pomegranate production in India has increased by 20–25% causing it to top the list of pomegranates producing countries. Earlier the scenario was different, Iran was the largest producer. Besides Iran, pomegranate grows in Turkey, Spain, China, Egypt. In India, Maharashtra contributes maximum to total production with 71% of

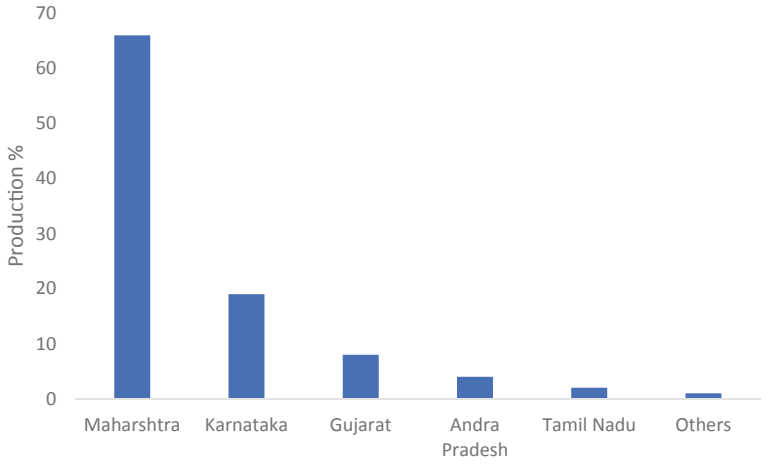



**Fig. 1** Production scenario of pomegranate in India (*Source* National Horticulture Board)

the total area under pomegranate cultivation in the country followed by Karnataka. Jammu and Kashmir too contribute significant proportion to the national pomegranate production Pomegranate cultivation is mainly concentrated in Ganderbal District of Kashmir Division with the total production of 2 MT of pomegranate from 13 hectares of land (National Horticulture Board) (Fig. 1).

Pomegranate is non-climacteric fruit and its consumption fruit dates to prophetic times. Pomegranate enjoy preference due to the deep red colour of arils and distinct shape. The arils of the pomegranate are consumed as fresh or in the form of frozen or dried product. However, because of hard, tough rind, consumer generally prefer the juice of the fruit. It is convenient and most of the therapeutic compounds are ingested via juice. Besides juice, pomegranates are processed into various value-added products like beverages, jams, and pastes. In addition, pomegranate is used as flavouring and colouring agent in various beverages and cuisines. Moreover, because of high sugar content, pomegranate juice makes best raw material for production of wine (Ghatge et al., 2005). Depending on juice yield, the cultivars are selected for extraction of juice, juice yield is given by the formula

$$\% \text{ juice} = (\text{juice wt} \times 100) + \text{whole fruit wt}$$

The two known varieties of pomegranate are *Punica granatum L.* and *P. protopunica*. *P. protopunica* is assumed to be ancestral species of *P. granatum L.* However, they are classified in different genera because of difference in habitats (Aulakh, 2004). Pomegranates trees are either evergreen or deciduous depending on the locality in which they grow. Pomegranates are deciduous in area of low-temperature (temperate) whereas are evergreen in sub-tropical areas. As the name suggests, there is no leaf fall in evergreen. Some commonly grown cultivars are Ganesh, Jyoti, Mridula, Ruby, P-16, P-23 and Yercaud-1 (Singh, 2009) (Figs. 2 and 3).



**Fig. 2** Pomagranate tree



**Fig. 3** Pomagranate

# 4 Nutritional Value and Chemical Composition

Pomegranate is a nutrient dense fruit with low calorific value in comparison to other fruits and is a source of all essential and easily digestible nutrients, which refreshes physically or mentally tired people. It is a diet full of nourishment for adolescents, ageing/elderly or sick persons and pregnant women. It has the power to cure hunger and quench thirst (Kushwaha et al., 2013) (Table 1).

- Carbohydrate

Pomegranate is relatively a good source of carbohydrates. A 100 g of pomegranate arils provide around 18.7 g of carbohydrates that too in the form of sugars. Pomegranate is considered a high GI fruit. The total sugar content of Pomegranate is 10 g/100 g of fruit weight with glucose being the predominant (5 mg/100 g) followed by fructose. Sorbitol (4.9 mg/100 g) is also present in good amounts and is responsible for laxative effect (Singh & Sethi, 2003).

- Fibre

Pomegranate is a medium source of fibre with the average fibre content of fruit 4 g/100 g fibre of the fruit consists of non-digestible carbohydrates and lignans. The major fibre in fruit is pectin (1.4%). The soluable and insoluable dietary fibre content of pomegranate is around 0.64 and 0.18%, respectively (Melgarejo & Artes, 2000).

**Table 1** Chemical composition of fresh pomegranate fruit

| Constituents | Nutritive Value per 100 g |
|---|---|
| Moisture | 78.0% |
| Protein | 1.67% |
| Fat | 1.17% |
| Carbohydrates | 18.7% |
| Fibre | 4.0% |
| Calorific value | 83 kcal |
| Vitamin C | 10.2 mg |
| Thiamine | 0.06 mg |
| Riboflavin | 0.10 mg |
| Niacin | 0.3 mg |
| Nicotinic acid | 0.3 mg |
| Minerals | 0.7% |
| Iron | 0.3 mg |
| Phosphorous | 36 mg |
| Choline | 7.6 mg |
| Potassium | 236 mg |
| Sodium | 3 mg |

*Source* United States Department of Agriculture (USDA)

**Table 2** Amino acids composition of pomegranate

| Amino acid | Percentage (g/100 g) |
|---|---|
| Histidine | 0.22 |
| Aspartic acid | 0.30 |
| Glutamic | 0.52 |
| Serine | 0.11 |
| Glycine | 0.41 |
| Arginie | 0.23 |
| Alanine | 0.19 |
| Proline | 0.14 |
| Lysine | 0.19 |
| Leucine | 0.19 |
| Isoleucine | 0.1 |
| Valine | 0.14 |

*Source* Rowayshed et al. (2013)

- Protein

Unlike most of the fruits, pomegranate provides good amount of protein. The average protein content of fruit is 1.67%. Aspartic acid, Arginine and glutamine are the predominant amino acid in pomegranate followed by glutamic acid and histidine. However, other amino acids such as alanine, tryptophan, methionine, valine, phenylalanine, isoleucine, lysine are also present. In addition, leucine and glycine are present in trace elements (Kushwaha et al., 2013) (Table 2).

- Lipid

Pomegranate are low in fats (0.72 g/100 g of fruit weight). Both unsaturated (0.14 g) as well as saturated fats (0.19 g) are present in the fruit with saturated triglycerides being the predominant. The concentration of ω-3 (linoleic acid) and oleic acid in pomegranate arils is 0.050 g/100 g and 0.048 g/100 g, respectively (Fadavi et al., 2006).

- Vitamins and minerals

Pomegranate is an excellent source of fat soluble vitamin phylloquinone (vitamin K), anti-coagulating and anti-haemorrhagic vitamin and provide tocopherol (vitamin E), anti-sterility vitamin, too. B complex vitamins such as pantothenic acid (vitamin $B_5$) and folic acid (vitamin $B_9$) are also present in pomegranate in good quantities. In addition, thiamine, riboflavin and niacin are present in traces. Unlike most of the fruits, pomegranate is a relatively poor source of ascorbic acid (Kushwaha et al., 2013).

The rich mineral profile of pomegranate consists of magnesium, potassium, phosphorus, calcium and sodium. Potassium is the major element of the fruit whereas iron, zinc, manganese and selenium are also present. Selenium is an important trace

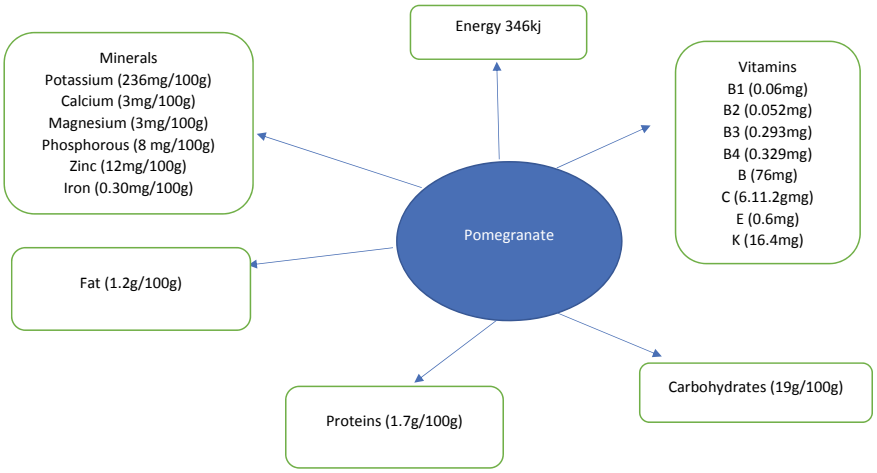



**Fig. 4** Chemical composition of pomegranate (*Source* Ullah et al., 2012)

element as it is a cofactor in the synthesis of selenoproteins. Moreover, copper is the trace mineral in pomegranate (Ullah et al., 2012) (Fig. 4).

- Phytochemicals

Over 100 phytochemicals have been identified in pomegranate fruit. Phenolic compounds of pomegranate include simple molecules (phenolic acids, phenyl-propanoids, tannins) or highly polymerized compounds (lignans, melanin, tannins). Pomegranate is considered a tannin-rich fruit and constitute hydrolyzable tannins such as punicalin, Peduneulagin and pumicalagin (Elfalleh et al., 2011).

Among all phenols, flavonoids, a 15-Carbon low molecular compound, is present in abundance by phenolic acid which is characterized by pressure of aromatic ring attached to one or more hydrogenated substitute. Flavonoids of pomegranate are luteolin and quercetin (Newman et al., 2007).

Anthocyanins, the largest flavonoid sub group, is responsible for colour of peel as well as skin. The major anthocyanins of Pomegranate are cyanidins (cyanidin-3-0-glucoside, cyaniding-3, 5-di-O glucoside, delphinidin (delphinidin-3-O-glucoside, delphinidin-3, 5,-di-O glucoside) and pelargonidin (pelargonidin-3-O-glucoside). Phenolic acids present in pomegranate are categorized as hydroxy benzoic acid (ellagic, gallic acid) and Hydroxy Cinnamic acid (caffeic, chlorogenic acid and P-coumaric acid) (Sharma & Maity, 2010).

## 5 Health Benefits

The bioactivities of compounds (including tannins, flavonoids, alkaloids, organic acids, triterpenes and steroids, etc. are responsible for pharmaceutical, pharmacological and medicinal benefits of pomegranate (Viuda-Martos et al., 2010a) (Fig. 5).

- Anti-diabetic effect

World Health Organisation has listed diabetes as third most pervasive disease after CVD and oncological disorders. Dietary interventions can play a major role in curbing rising levels of diabetes in population. Consuming pomegranate juice lowers blood glucose levels by blocking the activity of α-glucosidase (Goldenberg & Punthakee, 2013).

Pomegranate compounds with potential hypoglycaemic effect are oleanolic, ursolic and gallic acids. They limit the spread of diabetes by binding peroxisome proliferator-activated receptor γ (as ligands of this gene increase insulin sensitivity) and stabilizes nitric oxide production rates. Abnormal production of nitric oxide in body is linked with diabetes (Khalil, 2004). Animal model tests have showed that ethanolic extracts of pomegranate have potent hypoglycaemic activity in a dose dependent manner (Nogueira & Pereira, 1986).

Polyphenols of pomegranate exert hypoglycaemic activity by altering the blood glucose levels. Polyphenols inhibit the uptake of glucose by peripheral tissues via blocking the activity of gut glucosidase which is essential in glucose metabolism (Nogueira & Pereira, 1986).

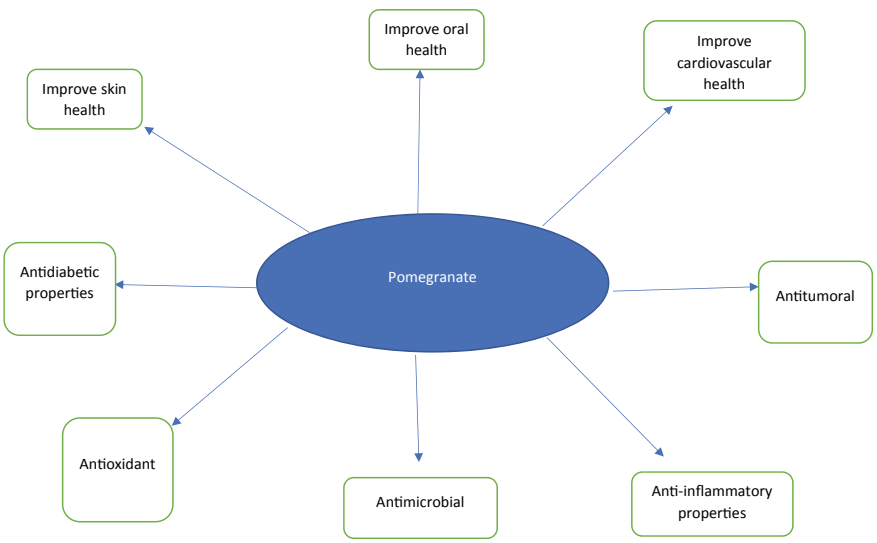



**Fig. 5** Potential health benefits of pomegranate (*Source* Viuda-Martos et al., 2010a)

- Improving skin health

Ageing of skin is a natural process, However certain factors accelerators the process, e.g. prolonged and exposure (UV radiations), oxidative stress and immune suppression. Polyphenols cause down regulation in formation of intracellular ROS and increase in intracellular antioxidant activity. Oral administration of pomegranate juice has whitening effect on skin due to impediment in proliferation of melanin inside melanocytes (Chidambara Murthy et al., 2004).

- Anti-microbial

Pomegranate has proven bactericidal effects against *Staphylococcus aureus*, *Proteus vulgaris*, *E. coli, Bacillus subtilis* and *salmonella typhi*. The antimicrobial effect of pomegranate fruit is attributed to its phenols, anthocyanins and tannins content. Gallic acid strong possesses strong inhibitory effect against *Salmonella enterica* via interfering in transcriptional steps of DNA formation. Different antimicrobials have different way of action. Some work by disrupting the cell membrane which cause leakage of vital cellular machinery while others pose hindrances in the activity of metabolic enzymes. Tannins are considered more toxic to microbes than the other two (phenols and anthocyanins). Tannins form stable complexes with proteins, carbohydrates, and metal ions (Fe and Cu) and these complexes manipulate the structure of enzyme which have a negative effect enzyme activity. Moreover, tannins make iron unavailable to bacterial cells by chelating the ions. Metal deficiency has deleterious effects on microbial cells. Phenols pose toxicity to microbes either by reacting with sulphydryl group of proteins thereby limiting the availability of substrate to microbes or by interfering in protein synthesis (Negi & Jayaprakasha, 2003).

In a study, application of pomegranate extract proved to be effective in inhibiting the growth and activity of *Staphylococcus aureus* even at low levels (Gibbons, 2004).

- Anti-inflammatory activity

The anti-inflammatory activity of pomegranate is believed to be due to punicic acid, a conjugated fatty acid. Consuming pomegranate causes inhibition of cyclooxygenase and lipoxygenase enzyme system by blocking the activation of neurophil, a type of white blood cell. Neurophil cells leads to inflammation and damage to tissue, joint, bones and cartilage via production of protease, cytokinis, chemokins, prostaglandins and ROS (Lansky & Newman, 2007).

Short-term inflammatory action of cells protect the body against physical injuries, poisons and wounds. However, continual inflammation results in disruption in normal physiology of body which in turn leads to asthma and arthritis. Low molecular weight compounds such as eicosanoids and enzymatic degradation of tissue along with several other factors cause inflammation. Cyclooxygenase, an enzyme which regulates haemostatics and facilitates arachidonic acid cycle, has been linked to anti-inflammatory action of pomegranate (Fan & Watanabe, 2003).

**Table 3** Chemical composition of different parts of pomegranate tree

| Parts | Chemical constituents |
|---|---|
| Seed | Oil (18%), punicic acid, alpha-estradiol, Icosanoic, Linolenic, Oleic, Palmitic, Stearic acid |
| Juice and peel | Sugars: fructose, sucrose, and glucose<br>Organic acids: ascorbic acid, citric acid, fumaric acid and malic acid<br>Amino acids: proline, methionine and valine<br>Polyphenols: tannins and flavonoids (Ellagitannin, anthocyanins, flavan-3-ols, and flavonols, catechins, gallic acid, protocatechuic acid, chlorogenic acid, caffeic acid, ferulic acid, o- and p-coumaric acids, catechin, phloridzin and quercetin |
| Bark and roots | Alkaloids |

*Source* Zarfeshany et al. (2014)

- Gastroprotective activity

*Halobacterium pyrolii* is the main cause of gastric ulceration. Gastric ulceration refers to the development of sole on the lining of stomach. The gastroprotective effect of pomegranate is due to the presence of its rich phytonutrients (saponins, tannins and flavonoids). Tannins are high molecular weight phenolic compounds which have the ability to form complexes with proteins present at the damaged site of epithelial tissue. The tannin-protein complex adheres to the stomach lining and acts as a protective layer on the damaged tissue thus, facilitating the healing process through preventing excessive bleeding. Moreover, pomegranate constituents (phenolics) directly prevent the gastric ulceration via altering the surface hydrophobicity of bacterial cell thereby, preventing the adhesion of these microbial cells to gastric lining (De Jesus et al., 2012).

Application of methanolic extract of pomegranate in a dose dependent to media containing *H. pyroli* has shown remarkable inhibition in growth and number of *pyrolii* cells (Repetto & Llesuy, 2002) (Table 3).

- Antioxidant

Repressive oxidative activity of pomegranate is contributed by phenolic compounds such as punicalagin, anthocyanins (cyanidin-3-glucosides, delphinidin) with punicalagin being the major antioxidant mostly concentrated in peels. These compounds undergo redox reactions as phenolic group can readily donate hydroxyl group (OH) in presence of reducing agents. In broader sense, phenolics manifest antioxidative action by scavenging free radicals or chelating metal ions (Negi et al., 2003).

Presence of reductones determine the reducing property of a compound. Reductones cause elimination of free radicals by liberation of hydrogen atom. They also interfere in peroxide formation by reacting with the precursor of peroxide ions and blocks the lipid peroxidation reaction.

**Table 4** Selected bioactive compounds in pomegranate evaluated with regard to prevention of diseases

| Pomegranate | Disease prevention activity |
|---|---|
| Ellagitannin | Antioxidant activity |
| Punicic acid | Anti-carcinogenic activity |
| Phenethyl rutinoside | Anti-diarroeal activities |

*Source* Kushwaha et al. (2013)

- Osteoarthritis and Chondroprotective activity

Osteoarthritis, degenerative arthritis, is a medical condition that results from damage to joints and the underlying bones. It generally prevails in elderly and the main etiology behind osteoarthritis are previous injury, inheritance factors or abnormal joint or bone development. Chondroprotection refers to a set of action which helps to retard, inhibit or repair degenerative joint injuries. Increased cytokine production and reduced growth factors control destruction and repair of osteoarthritis. Intake of pomegranate reduces the damage to cartilage in people suffering from arthritis. The fruit has the capability to reduce the inflammation and exert chondroprotective activity by inhibits the enzymes that destroy the cartilage. Animal model studies have shown that application of pomegranate extract reduced the onset and incidence of collagen-induced arthritis. Consuming pomegranate juice has been found to be effective in improving histopathological damages and exerting chondro protective effects (Rasheed et al., 2010) by repairing chondrocyte damage, causing no cell inflammation or proliferation in synovial fluid and limiting the damage caused to proteoglycan contents of cartilage tissue (Table 4).

- Protection against hypertension

Pomegranate juice exert inhibitory effect on serum angiotensin converting enzyme (ACE) activity as well as cytochrome pysoenzymes activity that facilitate "ACE" inhibition. ACE activity causes elevated blood pressure levels and accelerates the arte of atherosclerosis. Hence, intake of pomegranate juice reduces hypertension. The hypotensive action of pomegranate is attributed to its antioxidant profile such as tannins. Tannins are natural polyphenols, which are able to precipitate water-soluble alkaloids and exhibit anti hypertensive action by inhibiting the angiotensin converting enzyme (ACE).

- Protection against cardiovascular diseases

Elevated LDL and reduced HDL are the major etiology behind CVD occurence. Besides contributing to artheroseclorosis, LDL undergoes oxidation and causes deposition of fats on the artery walls. Oxidized LDL is more readily used by macrophages, white blood cells that remove cellular debris. Pomegranate maintains cardiovascular health by manipulating with metabolism of macrophage cholesterol via interfering with the uptake of oxidized LDL by macrophages and blocking the cholesterol biosynthesis. Both the steps eventually lead to decrease in macrophage

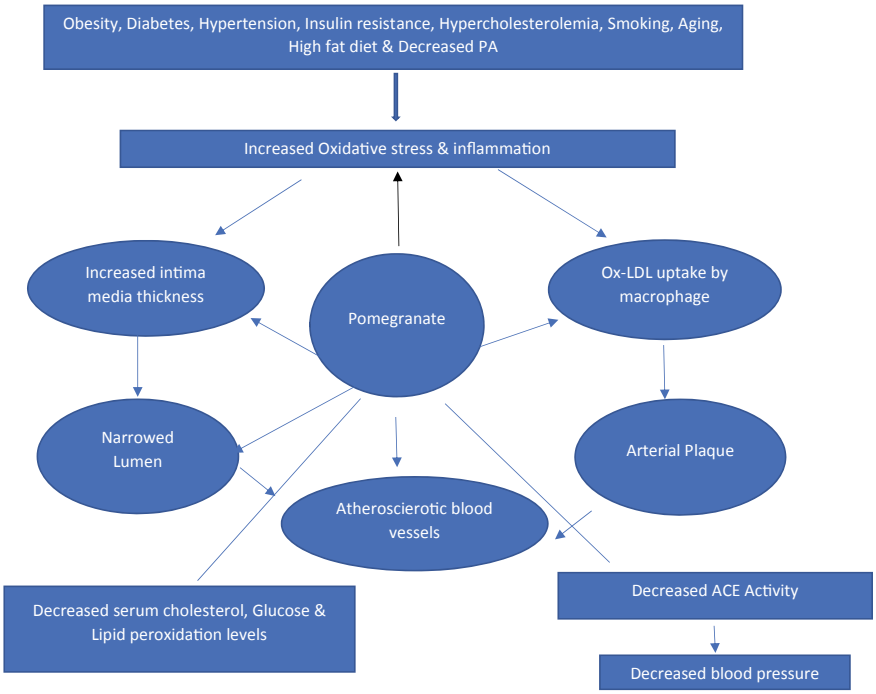



**Fig. 6** Mechanism depicting the cardioprotective effect of polyphenols of pomegranate (*Source* Viuda-Martos et al., 2010b)

cholesterol uptake and removal of deposited LDL plague. Moreover, consuming pomegranate juice inhibits ion (superoxide, peroxyl) which renders nitric oxide inactive (Viuda-Martos et al., 2010b) (Fig. 6).

- Anti cancer activity

The chemoprotective activity of pomegranate is believed to be due to upsurge in linolenic acid levels in mucosa and liver. Consuming pomegranate causes increased expression of PRAR γ gene (peroxime proliferators activated receptors), a non tumpor protein in mucosa. PRAR, present in adipose tissue, is essential for metabolism of fatty acids and glucose. Increased expression of this gene induces apoptosis in cancer cells. It is actually the ellagic acid, punicalagin and tannins of pomegranate which causes apoptosis in cancer cells. Other than these, caffeic acid, luteolin and punic acid too have proved to be effective in combating cancer by inhibiting the invasion of human PC-3 prostate cancer cell (Hora et al., 2003).

Anthocyanins too contribute to anticarcenogenicity of pomegranate by limiting the proliferation of colon cancer whereas ellagic acid prevent occurrence of colon cancer by altering the production of genes, associated with cancer development, in colon cell lining. In a study, administration of pomegranate seed oil inhibited the

development and propagation of colonic adenocarcinoma in testing subjects (Lansky & Newman, 2007).

## References


Ataseven Isık, E., & Celik, M. (2006). Similarities between pomegranate parents and crosses regarding some fruit characteristics. *I International Symposium on Pomegranate and Minor Mediterranean Fruits, 818,* 229–232.

Aulakh, P. S. (2004). Evaluation of pomegranate cultivars grown in the lower Shivalik's of Punjab. *Haryana Journal of Horticultural Sciences, 33,* 81–82.

Aulakh, P. S., & Sur, H. S. (1999). Effect of mulching on soil temperature, soil moisture, weed population, growth and yield in pomegranate. *Progressive Horticulture, 31*(3/4), 131–133.

Chidambara Murthy, K. N., Reddy, V. K., Veigas, J. M., & Murthy, U. D. (2004). Study on wound healing activity of Punica granatum peel. *Journal of Medicinal Food, 7*(2), 256–259.

De Jesus, N. Z. T., Falcao, H. D. S., Gomes, I. F., Leite, T. J. D. A., Lima, G. R. D. M., Barbosa-Filho, J. M., … Batista, L. M. (2012). Tannins, peptic ulcers and related mechanisms. *International Journal of Molecular Sciences, 13*(3), 3203–3228.

Elfalleh, W., Tlili, N., Nasri, N., Yahia, Y., Hannachi, H., Chaira, N., … Ferchichi, A. (2011). Antioxidant capacities of phenolic compounds and tocopherols from Tunisian pomegranate (Punica granatum) fruits. *Journal of Food Science, 76*(5), C707–C713.

Fadavi, A., Barzegar, M., & Azizi, M. H. (2006). Determination of fatty acids and total lipid content in oilseed of 25 pomegranates varieties grown in Iran. *Journal of Food Composition and Analysis, 19*(6–7), 676–680.

Fan, J., & Watanabe, T. (2003). Inflammatory reactions in the pathogenesis of atherosclerosis. *Journal of Atherosclerosis and Thrombosis, 10*(2), 63–71.

Ghatge, P. U., Kulkarni, D. N., Rodge, A. B., & Kshirsagar, R. B. (2005). Studies on post-harvest treatments for increasing storage life of pomegranate. *Journal of Soils and Crops, 15*(2), 319–322.

Gibbons, S. (2004). Anti-staphylococcal plant natural products. *Natural Product Reports, 21*(2), 263–277.

Goldenberg, R., & Punthakee, Z. (2013). Definition, classification and diagnosis of diabetes, prediabetes and metabolic syndrome. *Canadian Journal of Diabetes, 37,* S8–S11.

Hora, J. J., Maydew, E. R., Lansky, E. P., & Dwivedi, C. (2003). Chemopreventive effects of pomegranate seed oil on skin tumor development in CD1 mice. *Journal of Medicinal Food, 6*(3), 157–161.

Khalil, E. A. (2004). Antidiabetic effect of an aqueous extract of Pomegranate (Punica granatum L.) peels in normal and alloxan diabetic rats. *The Egyptian Journal of Hospital Medicine, 16*(1), 92–99.

Kushwaha, S. C., Bera, M. B., & Kumar, P. (2013). Nutritional composition of detanninated and fresh pomegranate peel powder. *IOSR Journal of Environmental Science, Toxicology and Food Technology, 7*(1), 38–42.

Lansky, E. P., & Newman, R. A. (2007). Punica granatum (pomegranate) and its potential for prevention and treatment of inflammation and cancer. *Journal of Ethnopharmacology, 109*(2), 177–206.

Melgarejo, P., & Artes, F. (2000). Total lipid content and fatty acid composition of oilseed from lesser known sweet pomegranate clones. *Journal of the Science of Food and Agriculture, 80*(10), 1452–1454.

Negi, P. S., & Jayaprakasha, G. K. (2003). Antioxidant and antibacterial activities of Punica granatum peel extracts. *Journal of Food Science, 68*(4), 1473–1477.

Negi, P. S., Jayaprakasha, G. K., & Jena, B. S. (2003). Antioxidant and antimutagenic activities of pomegranate peel extracts. *Food Chemistry, 80*(3), 393–397.

Newman, R. A., Lansky, E. P., & Block, M. L. (2007). A wealth of phytochemicals. In *Pomegranate: The most medicinal fruit*. Basic Health Publications.

Nogueira, D. G., & Pereira, N. A. (1986). Effect of an alcoholic extract from the pericarp of roma (Punica grantum L.) on insulin hypoglycaemia mice. *Review Brasial Farm, 67,* 59.

Rasheed, Z., Akhtar, N., & Haqqi, T. M. (2010). Pomegranate extract inhibits the interleukin-1β-induced activation of MKK-3, p38α-MAPK and transcription factor RUNX-2 in human osteoarthritis chondrocytes. *Arthritis Research & Therapy, 12*(5), 1–13.

Repetto, M. G., & Llesuy, S. F. (2002). Antioxidant properties of natural compounds used in popular medicine for gastric ulcers. *Brazilian Journal of Medical and Biological Research, 35*(5), 523–534.

Sharma, J., & Maity, A. (2010). Pomegranate phytochemicals: Nutraceutical and therapeutic values. *Fruit, Vegetable and Cereal Science and Biotechnology, 4*(2), 56–76.

Singh, A. (2009). *Fruit physiology and production*. Kalyani Publishers.

Singh, D., & Sethi, V. (2003). Screening of pomegranate genotypes for the preparation of quality grade anardana. *Journal of Food Science and Technology (Mysore), 40*(2), 236–238.

Ullah, N., Ali, J., Khan, F. A., Khurram, M., Hussain, A., Rahman, I. U., … & Ullah, S. (2012). Proximate composition, minerals content, antibacterial and antifungal activity evaluation of pomegranate (Punica granatum L.) peels powder. *Middle-East Journal of Scientific Research, 11*(3), 396–401.

Viuda-Martos, M., Fernández-López, J., & Pérez-Álvarez, J. A. (2010a). Pomegranate and its many functional components as related to human health: A review. *Comprehensive Reviews in Food Science and Food Safety, 9*(6), 635–654.

Viuda-Martos, M., López-Marcos, M. C., Fernández-López, J., Sendra, E., López-Vargas, J. H., & Pérez-Álvarez, J. A. (2010b). Role of fiber in cardiovascular diseases: A review. *Comprehensive Reviews in Food Science and Food Safety, 9*(2), 240–258.

Zarfeshany, A., Asgary, S., & Javanmard, S. H. (2014). Potent health effects of pomegranate. *Advanced Biomedical Research, 3,* 100.